Does insulin sensitivity vary with both glucose and total daily dose, and is there a clear relationship between the three?

Both dynamic ISF equations tested against 1,684 people in seven public study archives

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Summary

The question has three parts, and they do not receive the same answer.

Dose. Sensitivity does fall as total daily dose rises. The relationship appears in four independent measurements, one of which was free to find no relationship at all, and holding recent carbohydrate intake constant retains 83% of it. The exponent is -0.83 between people and -0.645 within a person, against the -1 that v1 assumes and the -2 that v2 assumes.

Glucose. Sensitivity varies with glucose, and it varies in the opposite direction to the one both equations encode. Measured without imposing a shape, sensitivity is 0.64 of its mid-range value below 120 mg/dL and rises to 1.16 by 190 to 300 mg/dL. v1 predicts that profile running from 1.18 down to 0.75, and v2 from 1.75 down to 0.54. Part of what remains after that correction is carbohydrate still absorbing: within four hours of a recorded meal, measured sensitivity is negative.

The relationship between all three. Both equations multiply a dose term by a glucose term and assume the two do not interact. Fitting the term where they meet gives -0.444 ($p = 3e-07$), roughly half the size of the dose term itself, so the assumption does not hold. The glucose profile is flatter for people on under 25 units a day than for people on over 60.

Taken together: the dose axis holds a clear relationship, the glucose axis holds one that is real and pointed the other way, and the two do not combine as either equation has them combine. Asked to predict the overnight fall, a static 1800 rule gave the lowest error at 34.1 mg/dL, v1 cost 2.6 mg/dL more, and v2 gave 140.9 mg/dL.

Background

Both dynamic ISF equations recalculate a person's sensitivity factor from their recent total daily dose, several times an hour, in place of the fixed value in their settings. They disagree about how steeply sensitivity falls as dose rises. The original, referred to throughout as v1, has sensitivity inversely proportional to daily dose. The revised form, v2, has it inversely proportional to the square of daily dose. For two people whose doses differ threefold, v1 implies a threefold difference in sensitivity and v2 a ninefold one. The two curves cross near 64 units a day, so which of them gives the stronger correction depends on which side of that a person sits.

I examined this question previously using data from people running loops, and concluded that the dose relationship was real but shallower than either equation assumed. That work left the question open for reasons that were structural. The cohort was a few hundred people, almost all on doses between 20 and 80 units a day, which gave little leverage either side of the crossover. Every estimate depended on the loop's own insulin model, which is the object under examination. And no open-loop comparison was available.

The public study archives address all of these. The analysis below uses seven of them, covering approximately 1,700 people, ages 2 to 82, daily doses from 7 to 107 units, and four automated systems alongside one cohort using none.

These equations derive from the same rule of thumb that most pump settings come from, and were built by people contributing their time to open-source automated insulin delivery. One finding here is that the rule holds up well against what people run. The finding I could not support is the squared form.

The archives

Study	System in use	People	Carbohydrate logged	Median days	Median dose
Loop	Do-it-yourself Loop, fixed sensitivity	842	yes	397	38.6 U
		196	yes	246	41.9 U

Study	System in use	People	Carbohydrate logged	Median days	Median dose
REPLACE-BG	Pump and sensor, no automation				
DCLP3	Control-IQ, adults and adolescents	112	no	183	50.0 U
DCLP5	Control-IQ, ages 6 to 13	100	no	203	37.4 U
PEDAP	Control-IQ, ages 2 to 5	98	no	273	13.6 U
IOBP2	Bionic pancreas	336	no	93	49.0 U

FLAIR is present in the archive but its release ships no pump file, so it contributes daily totals and nothing further.

Two rows carry disproportionate weight in what follows. REPLACE-BG ran in 2015 on a pump and a sensor with no algorithm between them, making it the only cohort where the insulin a person received was not selected by a system observing their glucose. Loop and REPLACE-BG are also the only cohorts where people recorded what they ate, so they are the only two where a meal can be screened out rather than inferred.

The extraction produced 2.2 million candidate overnight windows, of which 708,106 passed screening, from 1,659 people contributing at least forty each.

Method

None of these archives holds an insulin-on-board figure, a loop prediction, or in most cases a sensitivity setting. Everything was reconstructed from delivered basal and boluses.

Each window covers four hours from a start on the hour between 23:00 and 03:00. The fall in glucose across those four hours was measured against the insulin acting during them, obtained by convolving the delivery record with an insulin model. Loop subjects were given the models Loop itself ships, taken from the LoopKit source rather than approximated; the remaining cohorts were given the oref exponential. Dividing the fall by the acting insulin, holding starting glucose and prior glucose constant, yields milligrams per decilitre per unit.

Two aspects of that construction do most of the work.

Insulin was separated by when it was committed. Insulin already present when a window opened was determined before anything in that window occurred, so it cannot be a response to the glucose the window subsequently measures. Insulin delivered during the window can be, and under an automated system usually is. The two enter every fit as separate terms.

Screening looked only backwards. A window was retained or dropped on its starting glucose, on sensor coverage, and on time since the person last ate. It was never screened on the subsequent glucose trajectory. Excluding nights on which glucose rose, on the reasoning that something must have been eaten, removes disproportionately the nights on which insulin acted least effectively, which inflates the estimated sensitivity rather than cleaning it. Those filters exist in the code as a sensitivity arm and move nothing material.

Validation of the dose reconstruction

Loop recorded its own insulin-on-board figure at every bolus. That is the only point in these archives where an application's internal state was written down, so it can be recomputed from the pump record and compared against what the application held.

Across 158 people with sufficient records, the reconstruction tracked Loop's own figure at a median correlation of 0.927, with a median absolute difference of 0.32 units. This validates the dose reconstruction against a recorded quantity rather than against internal assumptions.

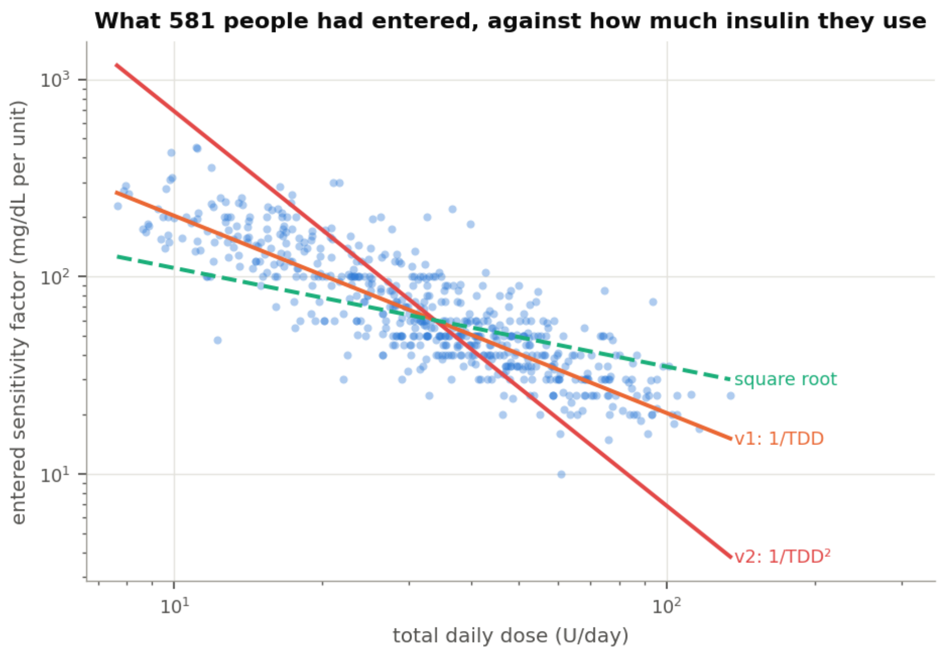
Producing it required something the archives do not obviously contain. Loop counts insulin on board net of scheduled basal, so a temporary basal contributes only the difference between what it delivered and what the profile would have delivered. Reconstructing that requires the basal profile, which appears in no settings file in the archive. It proved recoverable: every temporary basal record carries the rate it suppressed. Six and a half gigabytes of raw text reduces to 1.6 million rate changes, giving basal schedules for 842 people.

The limit of that check should be stated. Every candidate insulin model shows a small positive level bias, so the fastest-decaying model wins by absorbing it, which leaves the model each person was running only weakly identified. The correlation is the finding. The model selection is a weaker claim, and nothing downstream depends on it.

Settings people had entered

Of the 596 people with a recorded sensitivity factor, 581 also have thirty or more complete days of pump record from which to take a daily dose.

Their entered settings scale with daily dose at a log-log slope of -0.979, with a 95% interval from -1.030 to -0.934. Multiplying each person's sensitivity by their daily dose gives a median of 2067, against the 2139 that v1 evaluates to at a normal target. Under the 1800 rule that product should not vary with dose, and it does not: the Spearman correlation between them is +0.024. The equivalent test for the squared law returns +0.859, which indicates the relationship failing consistently across the range rather than at one end of it.



Entered sensitivity against daily dose

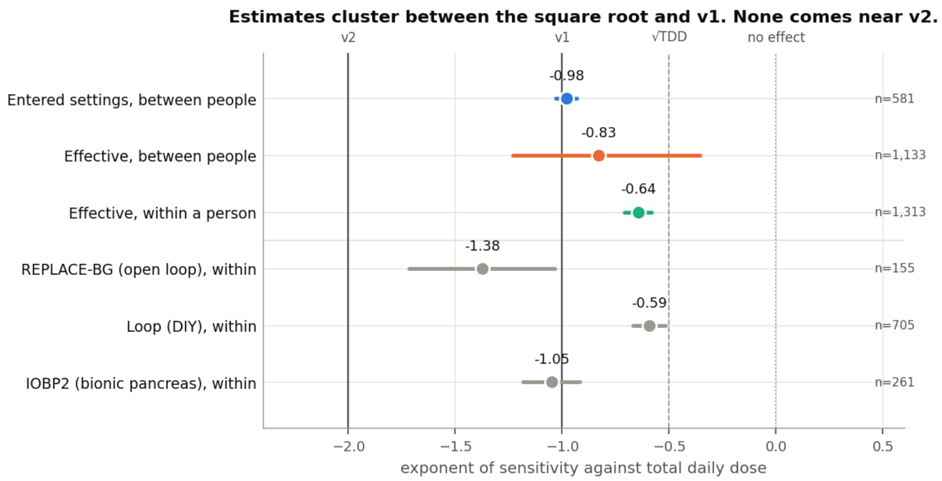
This result carries a caveat that should be stated rather than buried, because it approaches circularity. An entered sensitivity factor records a decision. People and their clinical teams commonly arrive at it through the 1800 rule, which is the source of v1's exponent. Finding that entered settings sit on 1800 over dose is therefore in part a finding about the reach of that rule in practice.

The circularity does not run in the other direction. Nothing in clinical practice directs people towards a squared law, and nothing in what people run resembles one.

Two secondary results are worth recording. Within a single age band the slope is shallower, between -0.68 and -1.10, so part of the pooled -0.98 reflects combining young children with adults. Carbohydrate ratios scale at -0.63 with a median product of 411, making the 500 rule a looser description of practice than the 1800 rule.

The dose relationship in the glucose response

The harder question is what a unit of insulin achieved, rather than what people had entered.



Exponents measured by each method

Between people, using only insulin committed before each window opened, the slope is -0.83 with a 95% interval of -1.23 to -0.36. Taking logarithms requires dropping anyone whose estimate came out negative, and those people are not a random subset, so the power law was also fitted directly on the natural scale where all of them survive. That fit returns -0.75.

Within a single person the exponent is -0.645, with a standard error of 0.033, pooled across 1,313 people, of whom roughly three quarters share the sign. This is the version that has not previously been tested, and it is the version these equations implement: they run inside one person, several times an hour, from a dose figure blended over recent hours and days. A relationship holding across a population need not hold within an individual.

Both estimates fall between the square root and v1. Neither approaches v2.

A second estimate that assumes no functional form

The measurements above fit a shape and report its parameter, which makes each answer conditional on the shape chosen. A gradient boosted model imposes none. Given 687,067 windows from 1,660 people it was free to find sensitivity falling with dose, rising with it, moving in steps, or not depending on dose.

The quantity to read from it is the interaction between insulin action and daily dose. Dose on its own has a substantial effect on the overnight fall that has nothing to do with sensitivity, because people on more insulin differ in other ways. Sensitivity is the multiplier on insulin, so the interaction is where it appears.

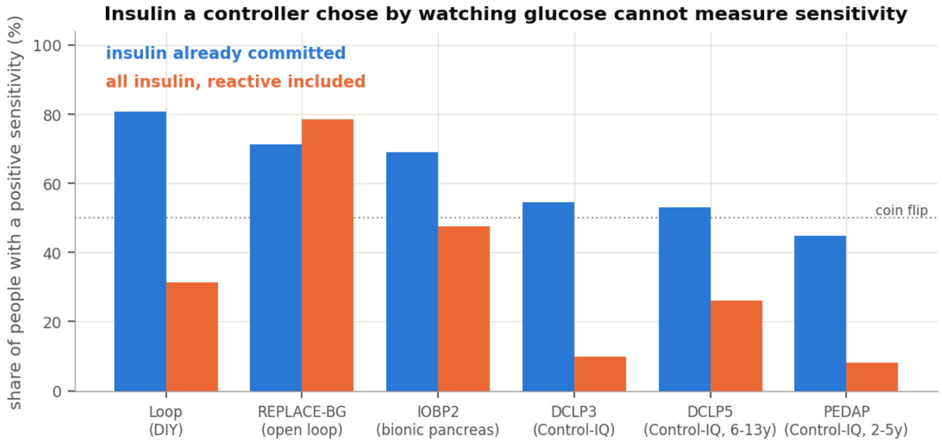
Daily dose	Shift in sensitivity attributable to dose
15 U	+0.28 mg/dL per unit
25 U	+0.06
34 U	+0.06
43 U	+0.02
55 U	0.00
77 U	-0.04

The profile declines monotonically without having been asked to, and flattens above roughly forty units a day, which is the shape a shallow exponent produces. The magnitudes should not be read as sensitivity itself. They are shifts around the model's own baseline and carry the same attenuation as every other estimate here.

Confounding by indication, and why one small cohort matters

There is a reason to distrust all of the above, and it is better demonstrated than argued.

Under an automated system, insulin approximates a function of recent glucose. More of it is delivered precisely when glucose is not falling. Regressing the fall on all the insulin that acted returns a negative slope, which would imply that insulin raises glucose. What that estimate measures is the algorithm's policy rather than anyone's physiology.



The confounding, made visible

Across the cohorts, the proportion of people whose sensitivity estimate is positive falls from 78% to 8% as the system becomes more reactive, and recovers when only committed insulin is used. REPLACE-BG behaves as the design predicts. It is the only cohort in which including all the insulin improves the estimate rather than degrading it, because in 2015 no algorithm was choosing it.

That is why 196 people from a decade ago carry weight out of proportion to their number, and why they are reported separately throughout. Their within-person exponent is -1.38, steeper than the pooled figure and the closest any cohort comes to v1. It does not approach v2 either.

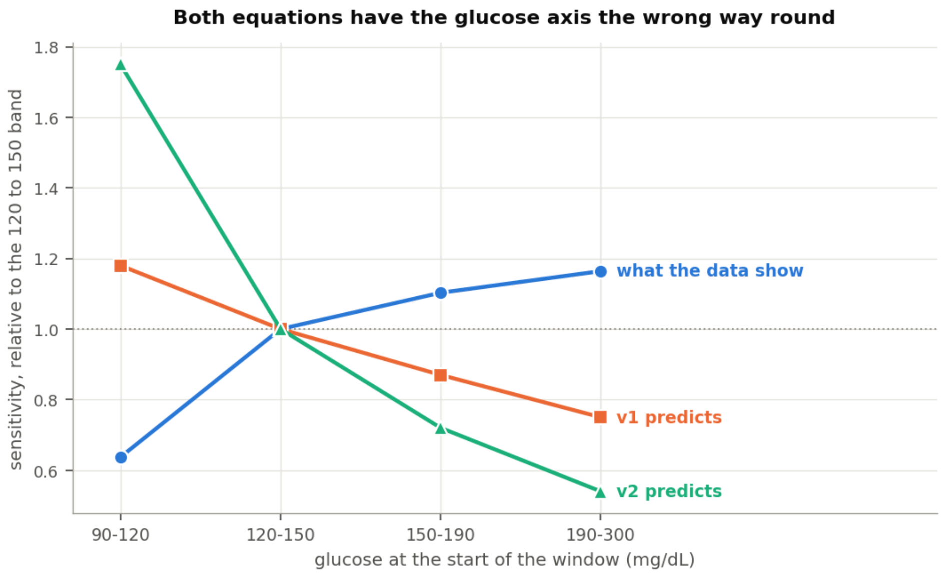
Dependence on glucose level

Both equations include a term that reduces sensitivity when glucose is high.

Testing that term has a trap in it. High glucose falls further than low glucose irrespective of insulin, through mass action, renal clearance and regression to the mean, and near target the body opposes a further fall. A model that allows glucose to explain the size of the fall, and then asks whether the fall per unit also depends on glucose, will answer the first question and record it as the second. Every fit here therefore carries an additive glucose term, and the claim rests on the interaction between insulin action and glucose, which is the only quantity implying that sensitivity itself moved.

The first attempt fitted a power law, as both equations do in their different ways. Pooled across 973 people the exponent came to +0.238 (SE 0.044), nominally in the direction the equations assume but around two fifths of what v1 encodes. It was also unstable. Restricting to windows starting between 120 and 220 mg/dL moved it to approximately +1.0, a shift larger than the quantity itself.

That instability had a cause worth more than the exponent. Repeating the fit with insulin action interacted with glucose band indicators instead of with a logarithm, on identical rows and with identical controls, gives a profile that no power law describes.



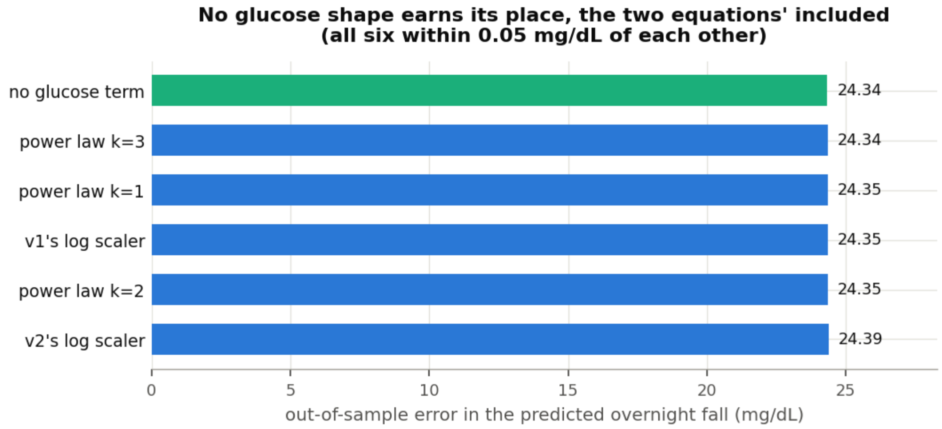
Sensitivity by glucose band against both equations

Glucose at window start	Measured	v1 predicts	v2 predicts
90 to 120 mg/dL	0.64	1.18	1.75
120 to 150	1.00	1.00	1.00
150 to 190	1.10	0.87	0.72
190 to 300	1.16	0.75	0.54

Sensitivity is around a third lower below 120 mg/dL and close to flat from there to 300. Fitting a monotone curve across that shape averages a step into a slope of almost nothing, which is why the power-law exponent sits near zero and changes sign with the range selected.

The direction is the opposite of what both equations encode. They make a unit of insulin achieve most near target and least when glucose is high. The measurement has it achieving least near target, which is where the body is defending against a further fall, and slightly more when glucose is high. An algorithm following either equation corrects hardest exactly where the measurement says insulin is working best, and eases off where it says insulin is working least.

The practical result follows from the shape.



Comparison of glucose shapes

Out of sample, across 1,616 people, six candidate shapes fall within 0.05 mg/dL of each other and the flat shape ranks first. Both equations' scalers rank in the lower half. The threshold set before running this analysis was an improvement of 0.5 mg/dL before a glucose term could be said to earn its place. The largest observed improvement was under a tenth of that. A term pointed the wrong way, at a magnitude this small, costs little and returns nothing.

An earlier draft of this document reported -2.67 for the power-law exponent. That was an indexing defect: the interaction term had been inserted ahead of the insulin coefficient in the design matrix while the coefficient was still read from a fixed position, so the interaction was divided by the starting glucose term. The error was found when a second implementation, written for the carbohydrate tests below, disagreed on the sign. Design matrix columns are now addressed by name and a regression test covers it. The shape comparison above uses a separate code path and was unaffected.

Whether the three combine as the equations assume

Both equations are separable. Written out, each asserts

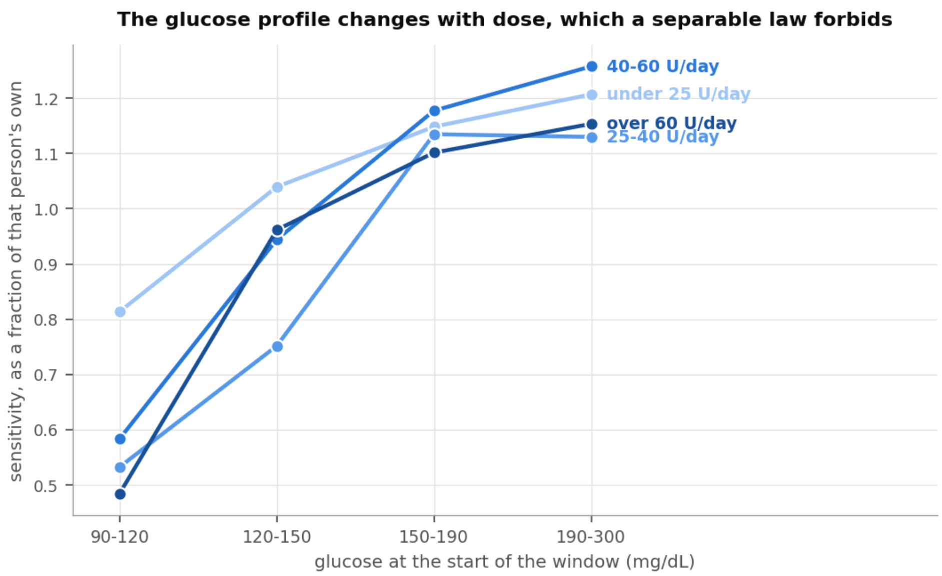
```
sensitivity = f(dose) x g(glucose)
```

with no term in which the two axes meet, so the glucose profile is the same whatever dose a person is on. That is a strong assumption and it had not been tested.

Fitting insulin action against glucose, against dose, and against the product of the two gives the interaction directly. Both equations predict it is zero.

Cohort	Glucose term	Dose term	Where they meet
Loop	+0.123	-0.747	-0.386 (p = 9e-05)
REPLACE-BG	+0.676	-1.217	-0.562 (p = 0.02)
IOBP2	+0.083	-2.033	-1.275 (p = 8e-08)
Pooled, 971 people	+0.140	-0.841	-0.444 (p = 3e-07)

The interaction is around half the size of the dose term. Measuring the glucose profile separately within each dose band shows the same thing without the parameterisation.



Glucose profile at each dose band

Daily dose	90 to 120	120 to 150	150 to 190	190 to 300
Under 25 U	0.81	1.04	1.15	1.21
25 to 40 U	0.53	0.75	1.14	1.13
40 to 60 U	0.58	0.94	1.18	1.26
Over 60 U	0.48	0.96	1.10	1.15

A separable law requires those rows to be identical. The near-target reduction is mild for people on under 25 units a day and around twice as deep for everyone else, so the glucose profile depends on the dose.

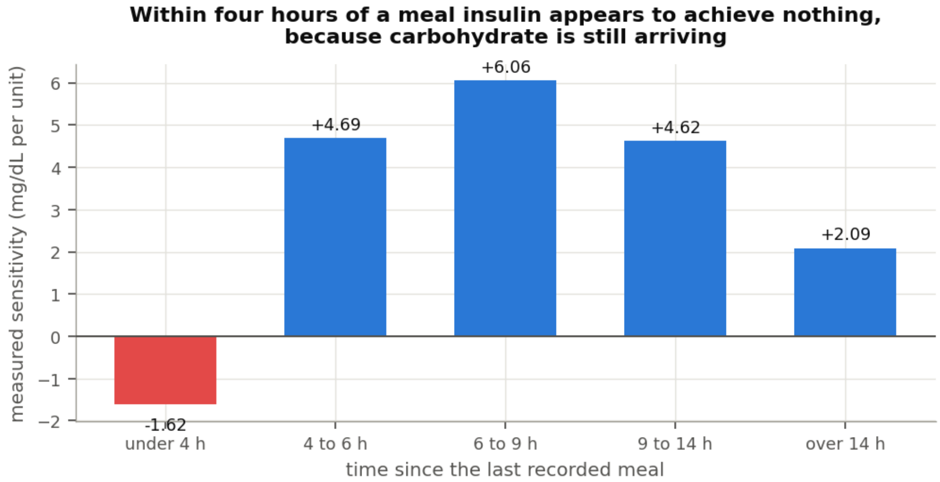
One thing does behave as the equations assume. A person's own glucose exponent shows no relationship with how much insulin they use (Spearman -0.017, n = 971), so the between-person version of separability survives even though the within-person version does not.

Physiology, or carbohydrate the model has not caught?

Both explanations predict the same observation. If sensitivity genuinely falls when glucose is high and when recent dose is large, the equations describe something real. If instead carbohydrate from a recent meal is still absorbing and unaccounted for, glucose is high for that reason, insulin appears to achieve less than it should, and recent dose is large because the person ate. The measurement is identical and the implication is not.

Loop and REPLACE-BG recorded what people ate, so time since the last meal is known rather than inferred. That allows three tests, each with a prediction that differs between the two explanations.

Sensitivity against time since the last meal

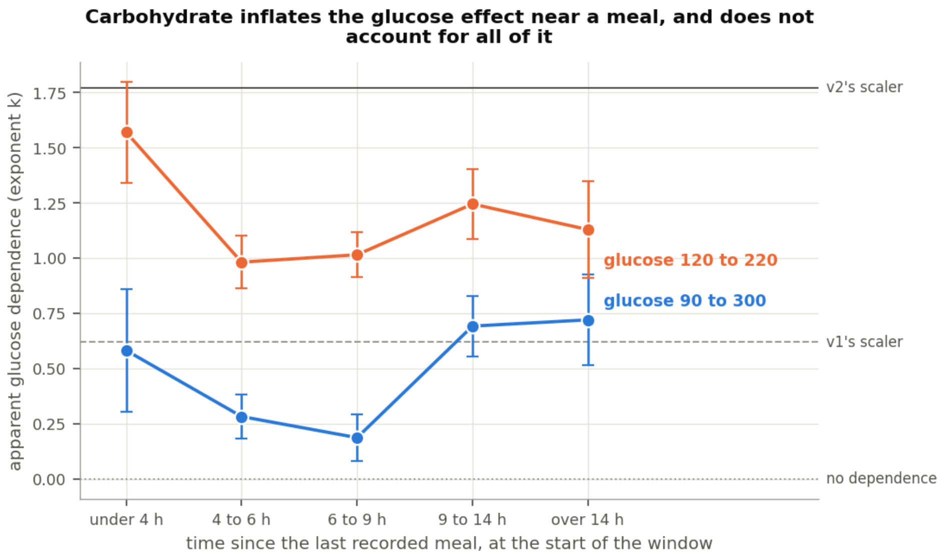


Measured sensitivity by time since the last meal

Within four hours of a recorded meal the median measured sensitivity is -1.62 mg/dL per unit. A negative value is not a physiological quantity. It states that glucose rose while insulin acted, which is what happens when carbohydrate is still arriving from a meal the absorption model has already discounted. The figure recovers to 6.06 mg/dL per unit in the six to nine hour band and declines again beyond that.

This is the absorption tail, measured directly, and it is large. It is also the reason the main screen requires six hours of clearance.

The apparent glucose dependence against the same axis



Apparent glucose dependence by time since the last meal

Under the carbohydrate explanation the apparent glucose dependence is strongest close to a meal and decays as absorption completes. Under the physiological explanation it is constant, because a person's sensitivity does not know when they last ate.

Time since last meal	Exponent, glucose 90 to 300	Exponent, glucose 120 to 220
Under 4 hours	+0.58	+1.57
4 to 6 hours	+0.28	+0.98
6 to 9 hours	+0.19	+1.01
9 to 14 hours	+0.69	+1.24
Over 14 hours	+0.72	+1.13
v1's scaler implies	+0.62	+0.62
v2's scaler implies	+1.77	+1.77

Close to a meal the exponent is roughly three times its value in clean fasting on the wider range, and around 55% larger on the narrower one. That part of the effect is carbohydrate.

It does not fall to zero. In the cleanest band the exponent remains positive and statistically distinguishable from zero, so carbohydrate does not account for all of the glucose dependence on this evidence.

The rise beyond nine hours has no explanation I am confident of. Windows more than fourteen hours clear of a recorded meal show a median of zero grams across the whole preceding day while carrying the second highest insulin on board of any band, which indicates food that was eaten and not recorded. Restricting to nights where the day's intake was recorded removes most of those windows and leaves the exponent unchanged at +0.82, so unrecorded food does not appear to be the explanation. It is reported here as an open question rather than resolved.

The dose relationship with recent carbohydrate held constant

A person whose recent dose is high has usually been eating, so the dose relationship could be the same artefact wearing different clothes. Adding grams recorded in the preceding 8 and 24 hours to the model, both directly and interacted with insulin action, tests that.

Cohort	Unadjusted	With carbohydrate held constant	Retained
Loop	-0.586 (SE 0.040)	-0.500 (SE 0.047)	85%
REPLACE-BG	-1.443 (SE 0.186)	-0.978 (SE 0.180)	68%
Pooled	-0.648 (SE 0.040)	-0.539 (SE 0.045)	83%

Most of the dose relationship survives. Carbohydrate accounts for roughly a sixth of it in Loop and around a third in REPLACE-BG, and the remainder is not explained by what people ate.

What the three tests support

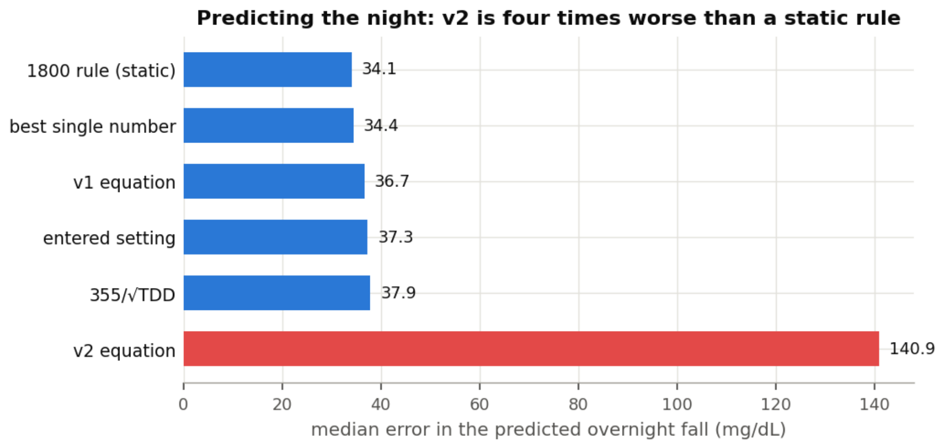
The dose half of the equations describes something that is mostly not carbohydrate. The glucose half is contaminated by carbohydrate to a degree that depends on how recently the person ate, is smaller than either equation encodes once meals are excluded, and varies by more than its own magnitude according to the glucose range it is measured over.

A fourth comparison, between cohorts that recorded meals and cohorts that did not, returned no consistent pattern and is not evidence either way. The cohorts differ in controller, in age and in dose alongside their record keeping, so the comparison was never well identified.

Predicting the overnight fall

The exponents address the science. The comparison below addresses the practical consequence of using either equation.

Each candidate received a per-person intercept fitted on that person's first 70% of nights and was scored on the remainder. This is generous to the equations. Most overnight insulin is basal, delivered to offset endogenous glucose production, so requiring the sensitivity factor to account for that offset would penalise an equation for a quantity it was never intended to supply.



Predicted overnight fall by candidate

Candidate	Median error	Median bias
Static 1800 rule	34.1 mg/dL	+2.9
Best single value for that person	34.4	-1.4
v1	36.7	+0.9
The person's own entered setting	37.3	-0.2
355 over the square root of dose	37.9	+3.8
v2	140.9	+41.1

Of 1,626 people, the best candidate was a single fitted value for 638, the static 1800 rule for 587, the square root form for 195, v1 for 135, and the person's own setting for 71. It was v2 for none of them.

v2's error is largest at the doses where the clinical consequence is greatest. For people on under twenty units a day the median error is 322 mg/dL, because squaring the dose gives a young child a sensitivity of several thousand milligrams per decilitre per unit and therefore a correction dose close to zero. The error falls as dose rises, reaching 83 mg/dL above 64 units a day, which is the crossover behaving as the algebra requires. Even there it is more than twice the error of any other candidate.

Recovering a known relationship

Every estimate above targets a quantity nobody observed, using a reconstruction of insulin that is imperfect. The fitted sensitivities fall well below what people had entered, so attenuation is present. What matters is whether that attenuation also bends the exponent. The way to establish this is to construct people whose sensitivity follows a chosen law and pass them through identical machinery.

Recovery of a known relationship

Simulated exponent	Recovered, no automation	With a reactive system	With 45% of meals unrecorded
-0.50	-0.50	-0.42	-0.72
-1.00	-1.02	-0.80	-1.03
-2.00	-1.57	-1.85	-1.56

Shallow exponents are recovered closely. Steep ones are compressed. The bottom row carries the argument: a squared law returns between -1.56 and -1.85 under every condition tested, including one in which nearly half of all meals go unrecorded and a third of a unit of error is applied to what acted. The measured range was -0.65 to -0.88.

Attenuation in the simulation runs between 0.38 and 0.63 of the true sensitivity, which is less severe than the observed gap against entered settings. The level is therefore not a quantity this method recovers, and no claim is made that it does. The exponent is.

Limitations

The level of sensitivity is not measured here. The fitted values are attenuated, so no constant for any equation can be read from this work. Only the shape of the relationship can.

The cohorts using automated systems are weak evidence in isolation. In the Control-IQ studies the proportion of people with a positive sensitivity estimate is close to chance, and for the youngest cohort it falls below it. Those cohorts replicate a direction rather than establishing one. The open-loop cohort and the entered settings carry the argument.

The analysis is confined to overnight fasting windows, so the carbohydrate tests above reach only into the tail of a meal rather than across the meal itself. The first four hours after eating are measured here and show sensitivity going negative, which bounds the size of the effect but does not model it. A proper treatment of the postprandial hours would need the absorption model itself under examination, which this work does not attempt.

Sensitivity varying with the day rather than with the dose falls outside this scope. Exercise, illness and a change in diet all move sensitivity over hours and days. Nothing here measures those effects, and nothing here argues against them.

Two faults in the underlying data were identified and repaired at the point of reading. Extended bolus durations arrive in milliseconds for three of the six studies and in seconds for the other two, so a one-hour square wave is recorded as a thousand hours; distributing a bolus across that interval placed insulin in 41,591 of one person's 46,176 five-minute periods. Separately, the database returns timestamps at microsecond resolution, which divided every basal total by a thousand until the reconstruction was checked against the database's own sums. Anyone else working from these archives will encounter both.

Interpretation

v1's assumption about dose is close to correct, and it is closest in the domain it was drawn from: it describes what people run their pumps at to within a few per cent. Measured against the glucose response it is somewhat steep, with the supported range falling between the square root and v1. Its glucose term does not earn its place, at little cost either way.

For v2 I found no support in these data. Entered settings run counter to it, the glucose response runs counter to it, it was not the best predictor for any of 1,626 people, and the simulation indicates that a squared law could not have presented as any of the measurements obtained. Its error is largest in children and in anyone on a small daily dose, who are least able to absorb a correction that is far too small.

The scope of that statement should be clear. It describes what a large and varied group of people did, and carries no implication about anyone's intent or competence. It is one analysis, confined to overnight fasting windows, drawn from archives never collected for this purpose. Work approaching the question by another route would be welcome, and the equations under examination advanced the discussion that this analysis rests on.

On the three-part question, the answers separate.

There is a clear relationship on the dose axis. It survives holding recent carbohydrate constant, it appears in a model given no functional form to fit, and its exponent lies between roughly -0.65 and -1.0.

There is a relationship on the glucose axis, and it points the other way. Both equations make insulin most effective near target. The measurement has it least effective there, by about a third, and close to flat across the rest of the range. Part of what the equations are responding to is carbohydrate still absorbing rather than a change in sensitivity.

There is no clear relationship between all three in the form either equation uses. The two axes interact, at roughly half the size of the dose term itself, so a separable product of a dose term and a glucose term cannot describe the surface. Whatever a defensible dynamic sensitivity would look like, it is not f(dose) multiplied by g(glucose).

Where a dose term is wanted, these data support an exponent between roughly -0.65 and -1.0, applied to a person's own level rather than used to set it. Given that a static 1800 rule out-predicted every dynamic form tested, the question I am left with concerns the placement rather than the exponent. Sensitivity is where these equations put the adjustment. The measurement that most needs it, on this evidence, is what a meal is still doing four hours after it was recorded, which is a carbohydrate absorption question wearing a sensitivity label.

Reproduction

The code is in `inv009/` in the `dynamic-isf-calculations` repository, with 28 tests. The window cache rebuilds in approximately nine minutes.

```
psql -d oref -f inv009/ingest/sql/05_settings.sql
python3 inv009/ingest/load_settings.py
python3 -m inv009.build_cache
python3 -m inv009.entered_isf && python3 -m inv009.effective_isf
python3 -m inv009.tdd_axis && python3 -m inv009.glucose_axis
python3 -m inv009.head_to_head && python3 -m inv009.synthetic
python3 -m inv009.loop_model_infer && python3 -m inv009.ml_shap
python3 -m inv009.carb_hypothesis && python3 -m
inv009.joint_surface
python3 -m inv009.figures
```

Windows run four hours from starts between 23:00 and 03:00, requiring 80% sensor coverage, a starting glucose between 90 and 300 mg/dL, and either six hours clear of logged carbohydrate or, where none is logged, four hours clear of a bolus large relative to that person's own daily dose. Per-person fits use ordinary least squares with a heteroskedasticity-robust standard error. Population figures pool the per-person estimates by the method of DerSimonian and Laird, as in the earlier work, so the two bodies of results can be read against each other.

Data attribution

The source of the data is the Loop Study (sponsored by the Jaeb Center for Health Research and funded by the Helmsley Charitable Trust), but the analyses, content and conclusions presented herein are solely the responsibility of the authors and have not been reviewed or approved by the study sponsor.

The source of the data is the Insulin Only Bionic Pancreas Pivotal Trial, but the analyses, content and conclusions presented herein are solely the responsibility of the authors and have not been reviewed or approved by the Bionic Pancreas Research Group or Beta Bionics.

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Dates in these archives are shifted or rebased per participant, so time of day is preserved and calendar date is not. Nothing here depends on calendar date.